GEORGE XUE

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EDUCATION

University of Toronto Toronto, Canada

BASc candidate in Engineering Science, Machine Intelligence + PEY Co-op; GPA: 3.72/4.00

Relevant coursework: Fundamentals of Deep Learning; Introduction to Robotics; Mathematics for Robotics.

RESEARCH INTERESTS

My research interests center on teaching foundation models physical knowledge and enabling them to apply that knowledge across diverse tasks and environments.

SKILLS

Programming: **Python**, **C++**

Machine learning: **Reinforcement Learning** (PPO, exploration, preference-based), **Computer Vision** (pose, retrieval, segmentation)

Platforms: **PyTorch**, **OpenCV**, **ROS1/ROS2**, **Isaac Sim**, NVIDIA Omniverse/OmniGraph, Unity, Git, Linux, MATLAB

Robotics: **Real-world robotic policy deployment** and **sim-to-real transfer**

Languages: English (fluent), Mandarin Chinese (fluent)

PROJECTS

La-JEPA: Compact Robot Policies with Latent World Modeling

Co-author; under review at ICLR 2027

- Compact world-action policy trained from scratch on **V-JEPA 2.1 features**, jointly predicting future representations and robot actions rather than reconstructing pixels.
- Two-frame **tube-motion encoding** captures recent motion; observation-only action attention enables action-only inference with cached visual keys and values.
- The La-JEPA L variant achieved **82.3% success on LIBERO-Plus** without embodied pretraining; physical evaluations covered four manipulation tasks.

Faster-WAM: Do World Action Models Need Deep Action Modules?

Co-author; under review at ICLR 2027

- World-model-centric policy with a **single-layer action expert** that accesses pretrained video representations across backbone layers through **Lite KV-Fusion**.
- Action expert is conditioned through world-model representations with aligned video/action positions; **cached visual context** reduces repeated inference computation.
- Achieved **78.3% success on LIBERO-Plus** using the same LIBERO-trained policy under distribution shifts, without additional embodied pretraining.

Decodable Cues, Unreliable Execution

First author; Diagnosing TTT Memory in World-Action Models; submitted to CoRL 2026 Memory Workshop

- Diagnostic study of fast-weight memory in manipulation, distinguishing decodable information from its behavioral influence and reliable execution through controlled memory interventions.

Agentic Aerial Cinematography (ACDC) Toronto, Canada

Co-first author (equal contribution); accepted at CoRL 2026 Agentic Robotics Workshop

- Built **vision-language retrieval** to rank shot candidates from exploratory footage and initialize waypoint selection for camera planning.
- Implemented language-conditioned, preference-based **Bayesian optimization** to refine camera poses over 3D reconstructions under limited evaluation budgets.
- Integrated **ROS 2** with **Unity** and internal tooling to execute closed-loop trajectories and record flight footage.

Experience

Learning, Embodied Autonomy, and Forecasting (LEAF) Lab Toronto, Canada

Former Research Assistant *Nov 2024 - Jul 2026*

- Built an **agentic RL pipeline** connecting language task specifications to reward models and **PyTorch** training; standardized data schemas for instructions, rollouts, and preference labels.
- Automated **LLM/VLM** preference extraction over rollout videos for **preference-based RL** in low-feedback settings.
- Implemented **exploration** baselines in **Isaac Sim** and added logging to training and data-extraction pipelines; worked with **Prof. Nicholas Rhinehart** on prototype integration and evaluation.

UofT Autonomous Drone Racing Team Toronto, Canada

Research Team *Apr 2024 - Oct 2025*

- Led a five-person controls group; coordinated simulation, controls, and perception workstreams through reviews and interfaces.
- Developed a **PPO-based** low-level controller; applied residual RL, domain randomization, curriculum learning, and reward shaping for trajectory tracking under camera noise and latency.

Awards

Euclid Mathematics Contest (UWaterloo): Top 25% (2022)

American Mathematics Competition (AMC 12): AIME Qualifier (2022)

FASE Dean's Honor List (U of T): 2023 Fall, 2024 Winter, 2024 Fall